

N step transition probability:

$$P_{ij}(n) = \sum_{k=1}^m P_{ik}(n-1) P_{kj}(1), P_{ij}(2) = \sum_{k=1}^m P_{ik} P_{kj} \text{ \& } P_{ij}(3) = \sum_{k=1}^m \sum_{l=1}^m P_{jk} P_{kl} P_{lj}$$

N step transition probability matrix:

$$P^{(2)} = P * P \text{ \& } P^{(3)} = P^{(2)} * P \quad \text{Markov Chain Steady State distribution : } \pi_x = \lim_{h \rightarrow \infty} P_h(x).$$

When steady state distribution exists $\pi P = \pi$.

Binomial Process:

Table for k = 4

n=2		n=3		n=4		n=5	
F _r	p	F _r	p	F _r	p	F _r	p
.0	1.000	2	1.000	.0	1.000	5.7	.141
.6	.958	.6	.958	.3	.992	6.0	.105
1.2	.834	1.0	.910	.6	.928	6.3	.094
1.8	.792	1.8	.727	.9	.900	6.6	.077
2.4	.625	2.2	.608	1.2	.800	6.9	.068
3.0	.542	2.6	.524	1.5	.754	7.2	.054
3.6	.458	3.4	.446	1.8	.677	7.5	.052
4.2	.375	3.8	.342	2.1	.649	7.8	.036
4.8	.208	4.2	.300	2.4	.524	8.1	.033
5.4	.167	5.0	.207	2.7	.508	8.4	.019
6.0	.042	5.4	.175	3.0	.432	8.7	.014
		5.8	.148	3.3	.389	9.3	.012
		6.6	.075	3.6	.355	9.6	.0069
		7.0	.054	3.9	.324	9.9	.0062
		7.4	.033	4.5	.242	10.2	.0027
		8.2	.017	4.8	.200	10.8	.0016
		9.0	.0017	5.1	.190	11.1	.00094
				5.4	.158	12.0	.000072